Can a CT Scan Teach a Pathology Classifier? Cross-Dimensional Episodic Transfer from 3D Radiology to Few-Shot 2D Histopathology

Zulqarnain Ali[1]

[1]Department of Data Science, The Islamia University of Bahawalpur, Bahawalpur, Pakistan

Abstract

Deep learning for medical image analysis is bottlenecked by annotation cost, particularly in digital pathology, where expert labeling is slow and specialized. Meanwhile, radiological data such as computed tomography (CT) is comparatively abundant. We investigate whether self-supervised structural knowledge learned from 3D CT volumes can be transferred to improve few-shot classification of 2D multiplex pathology images, despite the two domains differing in dimensionality, modality, patient population, and anatomical site. We propose a framework that pretrains a multi-scale 3D encoder with contrastive self-supervision on unlabeled CT volumes, retrieves relevant CT context for each pathology sample via top-k cosine similarity against a frozen feature bank, fuses that context into pathology features through a cross-dimensional attention bridge with gated residual connections, aligns the resulting representation adversarially across modalities, and classifies pathology samples through an episodic prototypical network under N-way K-shot supervision. We evaluate the framework on a public radiology-to-pathology pair: 61 unlabeled thoracic CT volumes (LungCT-Diagnosis) as the source, and 268 weakly labeled head-and-neck squamous cell carcinoma (HNSCC) multiplex pathology samples from 42 patients (HNSCC-mIF-mIHC) as the target, with binary tumor-versus-non-tumor labels derived from tumor-mask coverage. Under grouped, patient-level three-fold cross-validation with $K \in \{2, 8\}$-shot adaptation, the framework achieves a mean accuracy of 88.66% (95% CI: 87.69–89.52), balanced accuracy of 88.26% (87.92–88.78), macro F1 of 0.884 (0.877–0.888), AUC-ROC of 0.970 (0.967–0.973), Cohen's κ of 0.769 (0.755–0.778), and MCC of 0.777 (0.760–0.786), with consistent performance across all three folds and a false-negative-dominated error pattern (mean sensitivity 81.2%, specificity 95.3%). A full-dataset inference pass with the best saved checkpoint reaches 94.40% accuracy on all 268 samples, though this pass is not a held-out evaluation. Despite strong discrimination, held-out calibration is poor (Expected Calibration Error $\approx$ 0.29), indicating the model's predicted probabilities overstate its true reliability – a finding we report explicitly rather than omit. Qualitative inspection of the learned embeddings shows pathology samples separating cleanly by class, while CT and pathology features remain visibly distinct by domain, indicating partial rather than complete representation-level alignment. We report these results as evidence of technical feasibility for a single, carefully validated source-target pair rather than as a general claim about cross-dimensional transfer; component-wise ablations, calibration correction, additional domain pairs, and external validation remain necessary future work.

1 Introduction

Deep learning has pushed automated diagnosis, segmentation, and quantification forward across nearly every medical imaging modality, but almost all of that progress rests on one assumption: that enough labeled data exists for the task at hand. That assumption breaks down quickly outside a handful of well-studied benchmarks. Rare cancer subtypes, specialized pathology

endpoints, and resource-limited clinical sites routinely lack the thousands of expert-annotated images that supervised deep learning expects, and the annotation bottleneck is structural rather than incidental: producing a single expert-labeled whole-slide image can take a pathologist tens of minutes to hours, and that expertise is itself scarce [1, 2].

At the same time, a different kind of medical imaging data is comparatively abundant. Computed tomography (CT) is acquired routinely for cancer staging, treatment planning, and follow-up, generating large volumes of three-dimensional radiological data as a byproduct of standard clinical care. This creates an asymmetry worth exploiting: if CT volumes encode tumor-relevant structural regularities that are not specific to the CT modality itself – boundary irregularity, tissue heterogeneity, density-driven texture – then a model pretrained on abundant CT data might supply a useful prior for a data-scarce pathology task, even though the two modalities differ in dimensionality (3D versus 2D), acquisition physics, and visual appearance.

This is the question this paper investigates: *can self-supervised structural knowledge learned from 3D CT volumes be transferred to improve few-shot classification of 2D multiplex pathology images, despite the mismatch in dimensionality between the two domains?* This differs from standard transfer learning and domain adaptation, both of which typically assume the source and target inputs already share a common dimensionality and only need their distributions aligned [3, 4]. Here the source domain is volumetric and the target domain is planar, so any transfer mechanism must first bridge a representational gap before it can address the usual concerns of distribution shift and limited supervision.

We study this question in a concrete clinical setting: transferring knowledge from lung CT volumes to the classification of head and neck squamous cell carcinoma (HNSCC) in multiplex immunofluorescence/immunohistochemistry (mIF/mIHC) pathology images. The task is binary tumor-versus-non-tumor classification, supervised with weak labels derived from tumor-mask coverage rather than a dedicated expert classification campaign – a realistic reflection of how labels are actually obtained when a large annotation study is not feasible. The two source and target datasets are otherwise unrelated: they come from different anatomical sites, different patients, and different imaging modalities. Any transfer benefit therefore has to come from generic tumor-relevant structure rather than from shared anatomy.

We propose a framework that combines four components: (1) a self-supervised 3D convolutional encoder pretrained with contrastive learning on unlabeled CT volumes, (2) a persistent CT feature bank with top-k cosine-similarity retrieval, (3) a cross-dimensional attention bridge that fuses retrieved CT context into pathology features through multi-head cross-attention and a learned gated residual, and (4) an episodic prototypical network trained under adversarial domain alignment for few-shot ($K \in \{2, 8\}$) pathology classification. The framework is evaluated with grouped, patient-level three-fold cross-validation on 268 pathology samples from 42 patients, using a CT source pool of 61 volumes that is never used with pathology labels and contributes only through self-supervised pretraining and feature retrieval.

Contributions. This paper makes the following contributions:

- A formulation of *cross-dimensional* knowledge transfer that is explicit about what changes when the source domain is volumetric and the target domain is planar, distinguishing it from same-dimensionality domain adaptation and from same-modality few-shot learning.
- A retrieval-aware transfer architecture that couples CT feature banking, a cross-dimensional attention module (CDAM), gated residual fusion, and adversarial alignment with episodic prototypical few-shot learning, evaluated end to end on a public radiology-to-pathology domain pair.
- An empirical study on grouped patient-level cross-validation showing that the transferred framework reaches 88.7% mean accuracy and 0.970 mean AUC-ROC on binary HNSCC

tumor classification under $K \in \{2, 8\}$-shot supervision, with tight cross-fold confidence intervals.

- A transparent discussion of what the reported evidence does and does not establish, including a case where direct inspection of the released visualizations and recomputation from released prediction files contradicts an optimistic reading (calibration, embedding alignment) and a case where the relevant experiments (component ablations, statistical significance testing) were simply not run (Section 7).

The remainder of the paper reviews related work in few-shot learning, domain adaptation, and cross-modal medical imaging (Section 2); describes the proposed architecture and training objective (Section 3); details the experimental setup, datasets, and reproducibility protocol (Section 4); reports quantitative and qualitative results (Section 5); and discusses findings, limitations, and future directions (Section 6–Section 8).

2 Related Work

Few-shot learning in medical imaging. Few-shot learning (FSL) addresses classification and segmentation when only a handful of labeled examples per class are available. Recent surveys of medical FSL organize the field into metric-based, optimization-based, and hybrid strategies, and consistently note that most evaluated tasks keep support and query examples within a single modality [1, 2]. Prototypical networks remain a popular metric-based choice due to their simplicity and strong empirical performance, and retrieval-augmented variants have recently been used to adapt segmentation foundation models to new medical tasks with a small support memory [5]. None of this work tests whether an episodic few-shot classifier can benefit from a support signal drawn from a structurally different, higher-dimensional modality, which is the setting studied here.

Domain adaptation and transfer learning. Domain adaptation methods reduce the distributional gap between a labeled source and an unlabeled or sparsely labeled target, commonly through adversarial feature alignment with a gradient-reversal layer [3, 4]. Surveys of this literature organize approaches into supervised, semi-supervised, adversarial, and discrepancy-based families, but they consistently assume that source and target inputs already occupy comparable input spaces, whether 3D-to-3D or 2D-to-2D [3]. Standard transfer learning in medical imaging typically pretrains on a large labeled or self-supervised corpus and fine-tunes on the target task, an approach now common in transformer- and foundation-model-based medical pipelines [6–9], but again within a fixed dimensionality. Adapting adversarial alignment to a setting where the "domain" difference is partly a dimensionality difference, not only a distributional one, has not, to our knowledge, been evaluated on a public radiology-pathology pair.

Multi-modal and cross-modal medical imaging. Multi-modal fusion has been explored extensively for modalities that already share structure, such as combining MRI sequences, or aligning radiology images with their text reports [10–15]. Surveys of multimodal cancer detection and multimodal computational pathology highlight the promise of combining imaging and non-imaging signals but also flag limited validation and a reliance on paired or directly comparable inputs [16–18]. The present setting is harder in one specific respect: the source (CT) and target (pathology) samples are not paired at the patient or lesion level, so any useful transfer must come from generic structural regularities rather than sample-level correspondence.

Foundation models and self-supervised pretraining in pathology and radiology. Self-supervised and weakly supervised learning have become central to reducing annotation

Table 1: Positioning of the present study relative to four related literature strands.

Strand	Typical assumption	What differs here
Few-shot medical imaging [1, 2, 5]	Support and query samples share one modality	Support-side prior is transferred from a structurally different, higher-dimensional modality
Domain adaptation [3, 4]	Source and target share input dimensionality	Source is 3D, target is 2D; alignment must act after an explicit dimensionality bridge
Multi-modal fusion [10, 16, 18]	Modalities are paired or directly comparable	CT and pathology samples are unpaired, from different patients and anatomical sites
Self-supervised / foundation pretraining [8, 21, 30]	Pretraining and fine-tuning occur in the same modality	Pretraining occurs on CT; the learned prior is queried, not fine-tuned, during pathology adaptation

cost in computational pathology, with contrastive and predictive pretraining shown to remain competitive when only a small number of labeled examples are available downstream [8, 19, 20]. Foundation-model benchmarking in pathology [21, 22] and radiology [9, 14, 15] shows rapid progress in feature extraction and multitask adaptation [23, 24], and general-purpose segmentation foundation models such as SAM have motivated a family of medical adaptations [25–29], alongside architectures purpose-built for volumetric context such as Swin UNETR and rigorously validated 3D segmentation pipelines such as nnU-Net [30, 31]. This body of work motivates several of our architectural choices (a multi-scale volumetric encoder, contrastive pretraining, attention-based fusion) but stays within a single modality; it does not test cross-dimensional transfer from a volumetric source to a planar target.

Clinical context. HNSCC is a heterogeneous cancer group with clinically significant variability in outcome, and prior work on this disease has largely used clinical or tabular variables for outcome prediction rather than transferred imaging priors [32]. We use HNSCC pathology classification as the target task precisely because it reflects a realistic low-annotation clinical scenario rather than a curated benchmark.

Scope of this review. Cross-dimensional and cross-modal correspondence learning has also been studied outside medical imaging, for example in matching 3D shapes or scenes to 2D views in computer vision and robotics. We do not review that broader literature here and restrict the comparison above to medical imaging, since that is the setting the datasets, labels, and clinical motivation of this study come from; a fuller positioning against the general cross-modal retrieval literature is left for future work.

Positioning. Table 1 summarizes how the four literature strands above relate to the present study. Individually, each strand offers a component this work reuses (metric-based few-shot classification, adversarial alignment, attention-based fusion, and self-supervised pretraining), but each has been developed and validated within a single dimensionality or modality. The contribution here is not a new component in isolation, but the integration and empirical test of these components in a setting where the source is volumetric, the target is planar, and no sample-level pairing exists between them.

3 Methodology

3.1 Problem formulation

Let the source domain be a set of 3D CT volumes $D_S = \{x_i^s\}_{i=1}^{N_S}$, $x_i^s \in \mathbb{R}^{D\times H\times W}$, used without pathology labels. Let the target domain be a set of 2D multiplex pathology images $D_T =$

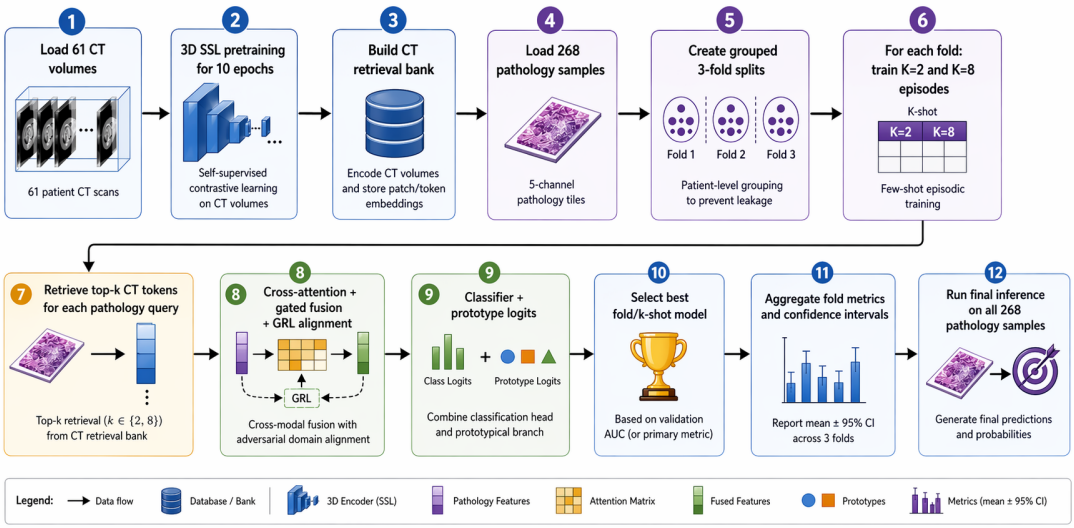


Figure 1: Two-phase training pipeline. Phase 1 pretrains a 3D CT encoder with contrastive self-supervision and builds a static CT feature bank. Phase 2 freezes the CT encoder in feature-bank mode and trains the cross-dimensional attention bridge, adversarial domain discriminator, and episodic prototypical classifier jointly on pathology support/query episodes.

$\{(x_j^t, y_j^t)\}_{j=1}^{N_T}$, $x_j^t \in \mathbb{R}^{C\times H'\times W'}$, $y_j^t \in \{0,1\}$ (non-tumor, tumor), with $N_T \ll N_S$ and N_T on the order of hundreds rather than the thousands of examples typically required for supervised deep learning. The goal is to learn a classifier for D_T that is more accurate and more data-efficient than one trained on D_T alone, by transferring structural knowledge from D_S despite the mismatch in dimensionality, channel count, and acquisition physics between the two domains. Unlike standard domain adaptation, D_S and D_T are not simply two distributions over a shared input space – they are different tensors entirely, so any transfer mechanism must first map both domains into a common representation before addressing distribution shift or limited supervision.

3.2 Overview

Figure 1 shows the two-phase pipeline. In Phase 1, a 3D encoder is pretrained with contrastive self-supervision on unlabeled CT volumes and then frozen; its outputs are cached in a CT feature bank. In Phase 2, the frozen CT bank is queried by top-k cosine similarity for each pathology sample, and the retrieved context is fused into the pathology representation through a cross-dimensional attention bridge, trained jointly with a domain discriminator (via a gradient-reversal layer) and an episodic prototypical classifier under N-way K-shot supervision, $N = 2$, $K \in \{2, 8\}$.

The framework has five components, shown in Figure 2: a 3D encoder E_{3D}, a 2D encoder E_{2D}, a retrieval-aware cross-dimensional attention bridge (CDAM), a domain discriminator D trained adversarially through a gradient-reversal layer, and a prototypical network for episodic few-shot classification.

3.3 Multi-scale 3D CT encoder

E_{3D} (`MultiScale3DEncoder`) takes volumes of shape $(1, 16, 128, 128)$ (channel, depth, height, width) and applies four convolutional stages with channel widths $32 \to 64 \to 128 \to 256$, each stage using two $3\times3\times3$ convolutions with batch normalization and ReLU, and max pooling after the first three stages. Global average pooling followed by a linear projection with LayerNorm

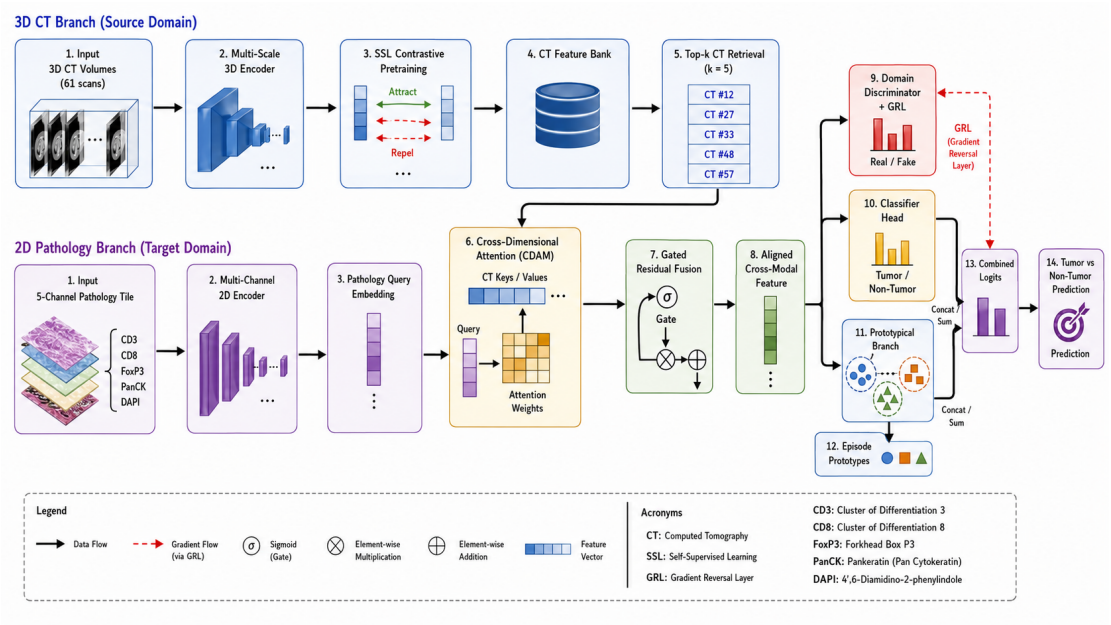


Figure 2: System architecture at inference time. A pathology query is encoded by the multi-channel 2D encoder, used to retrieve the top-k most similar CT volume-level descriptors from the pretrained feature bank, and fused with the retrieved context through the cross-dimensional attention module (CDAM) and gated residual connection before classification and prototype-based few-shot inference.

and ReLU maps the pooled 256-dimensional descriptor to a 512-dimensional feature $\mathbf{f}_{\mathrm{CT}}$:

$$\mathbf{f}_{\mathrm{CT}} = \mathrm{LayerNorm}(\mathrm{ReLU}(\mathbf{W}_{\mathrm{proj}} \cdot \mathrm{GlobalPool}(\mathbf{x}_4) + \mathbf{b}_{\mathrm{proj}})), \qquad \mathbf{W}_{\mathrm{proj}} \in \mathbb{R}^{512\times 256}. \tag{1}$$

A separate two-layer projection head maps $\mathbf{f}_{\mathrm{CT}}$ to a 256-dimensional embedding $\mathbf{z}_{\mathrm{CT}}$ used only for contrastive pretraining and discarded afterward.

3.4 Multi-channel 2D pathology encoder

E_{2D} (`MultiChannel2DEncoder`) processes the five pathology channels (CD3, CD8, FoxP3, PanCK, DAPI) with a dedicated initial 1→16-channel convolution per marker, producing per-channel descriptors $\mathbf{c}_1, \ldots, \mathbf{c}_5$. A channel-attention head pools each descriptor globally, concatenates the five pooled vectors, and passes them through a two-layer bottleneck with a sigmoid output to obtain per-channel weights $\mathbf{a} \in \mathbb{R}^5$, which rescale each channel's feature map before concatenation:

$$\mathbf{a} = \sigma(\mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\,\mathrm{concat}[\mathbf{g}_1, \ldots, \mathbf{g}_5])), \qquad \mathbf{x}_{\mathrm{weighted}} = \mathrm{concat}[a_1\mathbf{c}_1, \ldots, a_5\mathbf{c}_5]. \tag{2}$$

The weighted 80-channel representation then passes through four convolutional stages (32 → 64 → 128 → 256 channels) mirroring E_{3D}'s structure, followed by adaptive pooling and a projection to a 512-dimensional pathology feature $\mathbf{f}_{\mathrm{path}}$, with a matching 256-dimensional contrastive projection head $\mathbf{z}_{\mathrm{path}}$.

3.5 Top-k CT feature retrieval

The CT feature bank $\{\mathbf{c}_i\}_{i=1}^{N_{\mathrm{CT}}}$ stores exactly one global-pooled 512-dimensional descriptor $\mathbf{f}_{\mathrm{CT}}$ per CT volume ($N_{\mathrm{CT}} = 61$), produced by the same encoder and global-pooling operation described above; there is no sub-volume, slice-level, or patch-level tokenization. We refer to the k retrieved bank entries as "CT context vectors" below rather than "tokens" to avoid implying a finer retrieval granularity than what is actually implemented: top-k retrieval selects

k of the 61 whole-volume descriptors, not k local regions within one or more volumes. Before fusion, each pathology feature $\mathbf{q} = \mathbf{f}_{\text{path}}$ retrieves relevant CT context from this bank by cosine similarity, keeping the top $k = 5$ matches $\mathcal{T}_k$, softmax-reweighted with temperature $\tau = 0.1$:

$$\text{sim}(\mathbf{q}, \mathbf{c}_i) = \frac{\mathbf{q} \cdot \mathbf{c}_i}{\|\mathbf{q}\|\|\mathbf{c}_i\|}, \qquad w_i = \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{c}_i)/\tau)}{\sum_{j \in \mathcal{T}_k} \exp(\text{sim}(\mathbf{q}, \mathbf{c}_j)/\tau)}, \qquad \mathbf{r} = \sum_{i \in \mathcal{T}_k} w_i \mathbf{c}_i. \tag{3}$$

3.6 Cross-dimensional attention bridge (CDAM)

CDAM applies 8-head cross-attention between the pathology query and the k retrieved CT context vectors (key/value), followed by a feed-forward block with residual connections and layer normalization (hidden width 4×512, dropout 0.1):

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{f}_{\text{path}}, \quad \mathbf{K} = \mathbf{W}_K[\mathbf{c}_{\mathcal{T}_k(1)}, \ldots, \mathbf{c}_{\mathcal{T}_k(k)}], \quad \mathbf{V} = \mathbf{W}_V[\mathbf{c}_{\mathcal{T}_k(1)}, \ldots, \mathbf{c}_{\mathcal{T}_k(k)}], \tag{4}$$

$$\mathbf{f}_{\text{attn}} = \text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \qquad \mathbf{f}' = \text{LayerNorm}(\mathbf{Q} + \mathbf{f}_{\text{attn}}), \tag{5}$$

$$\mathbf{f}_{\text{FFN}} = \text{LayerNorm}(\mathbf{f}' + \text{MLP}(\mathbf{f}')). \tag{6}$$

A learned gate $\mathbf{g} = \sigma(\mathbf{W}_g \,\text{concat}[\mathbf{f}_{\text{path}}, \mathbf{f}_{\text{FFN}}, \mathbf{r}])$ controls how much of the attended CT context is injected into the pathology representation, combined with a directly adapted retrieval summary $\mathbf{r}_{\text{adapted}} = \text{GELU}(\text{LayerNorm}(\mathbf{W}_{\text{adapt}}\mathbf{r}))$:

$$\mathbf{f}_{\text{aligned}} = \mathbf{f}_{\text{path}} + \mathbf{g} \odot (\mathbf{f}_{\text{FFN}} - \mathbf{f}_{\text{path}}) + \lambda_{\text{res}}\, \mathbf{r}_{\text{adapted}}, \qquad \lambda_{\text{res}} = 0.20, \tag{7}$$

followed by a final LayerNorm over an output MLP, $\mathbf{f}_{\text{final}} = \text{LayerNorm}(\text{MLP}_{\text{out}}(\mathbf{f}_{\text{aligned}}))$. The gate is bounded away from zero (floor 0.15) so the transfer path cannot fully collapse to the identity mapping during optimization.

3.7 Adversarial domain alignment

A gradient-reversal layer (GRL) is the identity in the forward pass and negates the incoming gradient (scale $\alpha = 1.0$) in the backward pass, allowing a standard optimizer to play the adversarial minimax game between the feature extractors and a three-layer MLP domain discriminator D that predicts whether a feature originated from CT or pathology:

$$\min_{E_{3D}, E_{2D}} \max_{D} \; \mathbb{E}_{\mathbf{x} \sim P_{\text{CT}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim P_{\text{path}}}[\log(1 - D(\mathbf{x}))]. \tag{8}$$

This encourages $\mathbf{f}_{\text{final}}$ to be less separable by modality while the classification and prototypical losses (below) simultaneously push it to remain class-discriminative.

3.8 Episodic prototypical classification

For an N-way K-shot episode, the prototype for class n is the mean of its support-set features, and query samples are classified by softmax over negative squared Euclidean distance to the prototypes:

$$\mathbf{p}_n = \frac{1}{K} \sum_{i \in \mathcal{S}_n} \mathbf{f}_{\text{final},i}, \qquad p(y = n \mid \mathbf{f}_q) = \frac{\exp(-\|\mathbf{f}_q - \mathbf{p}_n\|^2)}{\sum_{n'} \exp(-\|\mathbf{f}_q - \mathbf{p}_{n'}\|^2)}. \tag{9}$$

Beyond episode-local prototypes, a global prototype per class is maintained with an exponential moving average, $\mathbf{p}_n^{\text{global}} \leftarrow \beta\, \mathbf{p}_n^{\text{global}} + (1 - \beta)\, \mathbf{p}_n^{\text{episode}}$, $\beta = 0.85$, so that prototype estimates stabilize across the small number of episodes seen from a 268-sample target dataset.

3.9 Training objective

The overall loss combines a main classification term with five auxiliary terms:

$$\mathcal{L} = \mathcal{L}_{\text{comb}} + 0.5\,\mathcal{L}_{\text{cls}} + \lambda_{\text{proto}}\mathcal{L}_{\text{proto}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{ctr}}\mathcal{L}_{\text{ctr}} + \lambda_{\text{consist}}\mathcal{L}_{\text{consist}} + \mathcal{L}_{\text{usage}}, \tag{10}$$

where $\mathcal{L}_{\text{comb}}$ is cross-entropy on a weighted combination of the classifier logits, the episodic-prototype logits, and the EMA global-prototype logits (the actual prediction path at inference time); $\mathcal{L}_{\text{cls}}$ is cross-entropy on the classifier logits alone, included as an auxiliary term with a fixed weight of 0.5; $\mathcal{L}_{\text{proto}}$ is the episodic prototypical negative log-likelihood from Eq. (9); $\mathcal{L}_{\text{adv}}$ is the adversarial domain-confusion loss from Eq. (8); $\mathcal{L}_{\text{ctr}}$ is a symmetric InfoNCE contrastive loss between the projection-head embeddings $\mathbf{z}_{\text{path}}$ and $\mathbf{z}_{\text{CT}}$, with positive and negative pairs defined as described below; and $\mathcal{L}_{\text{consist}}$ is the same combined-logit cross-entropy computed on the support set rather than the query set. $\mathcal{L}_{\text{usage}}$ is a transfer-usage penalty, computed separately for the query and support batches and summed, that discourages the gated fusion in Eq. (7) from collapsing onto the untransferred pathology features:

$$\mathcal{L}_{\text{usage}} = \lambda_{\text{pen}} \cdot \text{ReLU}(\text{sim}_{\text{raw}} - \text{sim}_{\text{aligned}} + \gamma) + \lambda_{\text{shift}} \cdot \text{ReLU}(\delta_{\min} - \|\mathbf{f}_{\text{aligned}} - \mathbf{f}_{\text{path}}\|), \tag{11}$$

where sim_{raw} and $\text{sim}_{\text{aligned}}$ are the cosine similarity between the retrieved CT features and, respectively, the raw and aligned pathology features, $\gamma = 0.05$ is a margin, and $\delta_{\min} = 0.03$ is a minimum required representation shift. Loss weights are $\lambda_{\text{proto}} = 0.5$, $\lambda_{\text{adv}} = 0.3$, $\lambda_{\text{ctr}} = 0.1$, $\lambda_{\text{consist}} = 0.1$, $\lambda_{\text{pen}} = 0.20$, and $\lambda_{\text{shift}} = 0.05$. We use distinct symbols λ_{ctr} and λ_{pen} here because, despite both being alignment-related weights, they multiply different loss terms in the implementation (the InfoNCE loss and the usage-penalty hinge, respectively) and should not be conflated. All hyperparameters are listed in Table 3.

How the InfoNCE pairs are actually defined. Because no ground-truth correspondence exists between a CT volume and a pathology sample, $\mathcal{L}_{\text{ctr}}$ cannot be defined against a known positive pair the way it would be for, e.g., paired image-report data. In the implementation, the "CT side" of the pair for a given pathology query is not an independently sampled CT example: it is $\mathbf{r}$, the same softmax-weighted combination of that query's own top-k retrieved bank vectors used in Eq. (7) (retrieval is computed on the detached query features, so no gradient flows back through the retrieval step itself). The InfoNCE loss then treats each query's projected $\mathbf{r}$ as its positive and the other examples' projected $\mathbf{r}$ vectors in the batch as negatives, pulling $\mathbf{z}_{\text{path}}$ toward a summary that was itself derived from that same query. This means $\mathcal{L}_{\text{ctr}}$ does not supply an alignment signal from an independent ground truth; it sharpens the model's embeddings to be more self-consistent with whatever the retrieval mechanism already ranked as relevant. We flag this as a genuine circularity risk rather than resolve it: the loss could in principle reinforce a self-consistent but semantically arbitrary retrieval pattern rather than a meaningful CT-pathology correspondence, and distinguishing these two possibilities would require the retrieval and ablation diagnostics listed in Section 7.

3.10 Training procedure

Phase 1 pretrains E_{3D} with contrastive self-supervision (InfoNCE on augmented CT-volume pairs, temperature 0.2) for 10 epochs, then freezes it and extracts the CT feature bank once from all 61 volumes. Phase 2 trains CDAM, D, and the prototypical classifier jointly for up to 30 epochs per K-shot setting with early stopping (patience 10), for each of the three grouped cross-validation folds and each $K \in \{2, 8\}$, and selects the best checkpoint per fold on the corresponding held-out validation split (Section 4). Algorithm 1 summarizes the full procedure.

Algorithm 1 Cross-dimensional transfer training pipeline

Pretrain E_{3D} with contrastive self-supervision on unlabeled CT volumes (Phase 1)
Freeze E_{3D}; extract and store the CT feature bank
for each grouped cross-validation fold $f \in \{1, 2, 3\}$ **do**
    for each support size $K \in \{2, 8\}$ **do**
        Sample pathology support/query episodes from the fold's training split
        Retrieve top-k CT context vectors per query; fuse via CDAM (Eq. (7))
        Update CDAM, D, classifier, and prototypes by minimizing $\mathcal{L}$
        Save the checkpoint with the best validation score for this (f, K)
    end for
    Evaluate the better of the two K-shot checkpoints on the fold's held-out test split
end for
Aggregate fold-wise metrics; run a full-dataset inference pass with the saved model

Table 2: Structural comparison of the source and target datasets.

Characteristic	**LungCT-Diagnosis**	**HNSCC-mIF-mIHC**
Modality	Computed tomography	Digital pathology (mIF/mIHC)
Dimensionality	3D volumetric	2D planar
Samples	61 volumes	268 samples (42 patients)
Native resolution	$512 \times 512 \times (102\text{–}401)$	$> 2000 \times 2000$ px
Channels	1 (grayscale, HU)	5 (multiplex markers)
Label usage	None (self-supervised only)	Binary, weak (tumor-mask ratio $> 5\%$)

4 Experimental Setup

4.1 Datasets

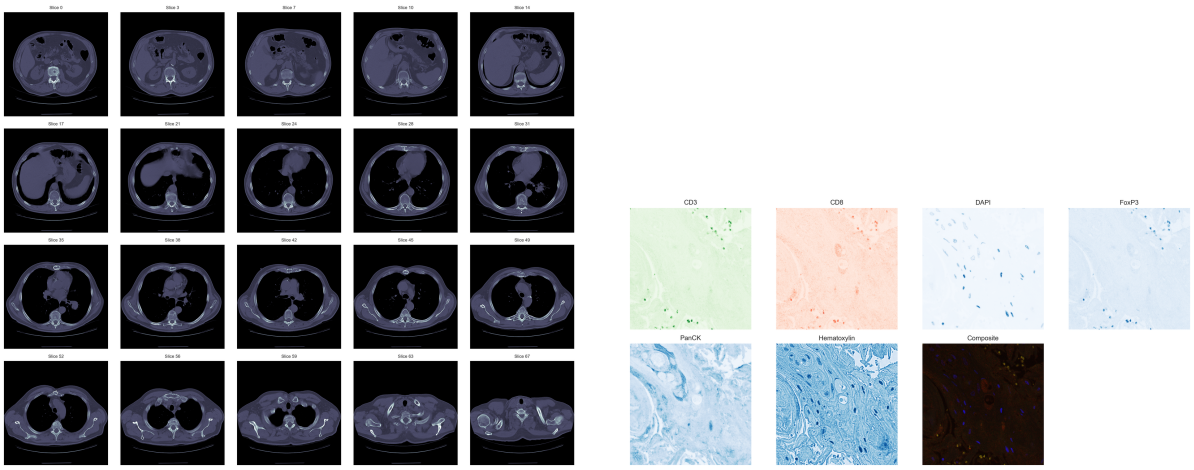
Source domain: LungCT-Diagnosis. 61 diagnostic, contrast-enhanced thoracic CT volumes from The Cancer Imaging Archive (TCIA)[1] (DOI: 10.7937/K9/TCIA.2015.A6V7JIWX), originally collected to study CT-derived shape-complexity and heterogeneity descriptors of lung adenocarcinoma. Volumes are 512×512 in-plane with 102–401 axial slices (mean 217.4) and slice thickness 3–6 mm. Diagnostic labels are not used; the collection serves purely as an unlabeled source for self-supervised pretraining and CT feature-bank construction.

Target domain: HNSCC-mIF-mIHC. 268 five-channel pathology samples (CD3, CD8, FoxP3, PanCK, DAPI) from 42 HNSCC patients, also from TCIA[2] (DOI: 10.7937/TCIA.2020.T90F-WB82), developed jointly by Memorial Sloan Kettering Cancer Center and Moffitt Cancer Center as a restained, co-registered mIF/mIHC comparison set. Nine regions of interest per patient (tumor core, tumor margin, adjacent stroma) are each subdivided into four 512×512 patches. Patients contribute 2–12 samples each (mean 6.4, median 6). Binary labels are derived from tumor-mask coverage: a sample is labeled tumor if the tumor-mask ratio exceeds 5%, giving a near-balanced split of 131 tumor (48.9%) and 137 non-tumor (51.1%) samples. Figure 3 shows representative examples from both domains; Table 2 summarizes their structural differences.

4.2 Preprocessing

CT volumes are windowed with lung settings (level -600 HU, width 1500 HU), sub-sampled every second slice, resized to 128×128 with bilinear interpolation, cropped/padded to a fixed

[1] https://www.cancerimagingarchive.net/collection/lungct-diagnosis/
[2] https://www.cancerimagingarchive.net/collection/hnscc-mif-mihc-comparison/



(a) LungCT-Diagnosis (source, 3D) **(b)** HNSCC-mIF-mIHC (target, 2D)

Figure 3: Representative examples from the source and target domains. The CT volumes (left) provide macroscopic, volumetric tissue-density information used only for self-supervised pretraining. The five-channel pathology samples (right) provide microscopic, planar molecular-marker information and carry the weak tumor/non-tumor labels used for supervision.

depth of 16 slices, and intensity-normalized to $[0, 1]$. Pathology channels are resized to 224×224 and normalized per channel with 2nd–98th percentile clipping to reduce staining and illumination variation, then stacked into a 5-channel tensor. Training-time augmentation uses random rotation, scaling, and flips for CT volumes, and random 90° rotation, flips, and color jitter for pathology images.

4.3 Cross-validation protocol

Because pathology samples are grouped by patient (2–12 samples per patient), a standard random k-fold split would leak correlated samples from the same patient across train and test partitions. We therefore use *grouped*, patient-level three-fold cross-validation: within each fold, all samples from a given patient fall entirely on one side of the split, approximately 67% of samples are used for training (with a further 15% held out for validation and model selection) and 33% for testing. The pretrained CT encoder and CT feature bank are held fixed across all folds and both K-shot settings, so only the pathology-side transfer components are re-trained per fold.

4.4 Hyperparameters

Table 3 lists the complete configuration used for the reported run, matching the `ResearchConfig` settings in the released implementation.

All experiments use a single fixed random seed (42) for Python, NumPy, and PyTorch; we report variability across the three grouped folds under this seed rather than across independent re-runs (Section 7). Training and evaluation were performed on a single NVIDIA Tesla P100 GPU; the full pipeline (Phase 1 pretraining, three grouped folds $\times$ two K-shot settings in Phase 2, and the final full-dataset inference pass) completed in approximately 71 minutes end to end, measured from the run's own timestamped log. The implementation uses Python 3.12 and PyTorch, with NumPy, SciPy, pandas, scikit-learn, matplotlib, seaborn, pydicom, PIL, and UMAP for data handling, evaluation, and visualization.

Table 3: Hyperparameter configuration.

Category	Parameter	Value
Data	CT volume size	$16 \times 128 \times 128$, 1 channel
	Pathology image size	224×224, 5 channels
Phase 1 (SSL)	Epochs	10
	Optimizer	AdamW, lr 10^{-4}, weight decay 10^{-4}
	Contrastive temperature	0.2
	Augmentation strength	0.5
Phase 2 (transfer)	Max epochs per $(fold, K)$	30
	Optimizer	AdamW, lr 10^{-4}, weight decay 10^{-4}
	LR schedule	Cosine annealing with warm restarts, 3-epoch warmup
	Early stopping	Patience 10, min. delta 10^{-4}
	Batch size / episodes per epoch	4 / 20–50 (adaptive)
Few-shot config	N-way	2
	K-shot	$\{2, 8\}$
	Query samples per episode	5
	Global prototype momentum β	0.85
CDAM	Top-k retrieval	5
	Retrieval temperature τ	0.1
	Attention heads	8
	Retrieval residual weight λ_{res}	0.20
Loss weights	λ_{proto} (prototypical)	0.5
	λ_{adv} (adversarial)	0.3
	λ_{ctr} (InfoNCE contrastive)	0.1
	λ_{consist} (support consistency)	0.1
	λ_{pen} (usage-penalty margin term)	0.20
	λ_{shift} (usage-penalty shift term)	0.05
Evaluation	CV folds (grouped, patient-level)	3
	Bootstrap resamples for CI	2000
	Random seed	42

4.5 Evaluation metrics

We report accuracy, balanced accuracy (mean of per-class recall, appropriate given that grouped folds do not guarantee identical class ratios), macro-averaged F1, AUC-ROC, Cohen's κ, and the Matthews correlation coefficient (MCC), which uses all four confusion-matrix entries and remains informative under class-ratio variation across folds. Fold-level means, standard deviations, and confidence intervals are reported over the three grouped folds.

5 Results

All numbers in this section are taken directly from the saved outputs of the reported run (`final_results.json`, per-fold `results.json` files, `experiment.log`, and the full-dataset inference summary), or are recomputed by us directly from those saved outputs using the released analysis script described in Section 4. We say explicitly where a number is a direct readout versus a quantity we derived.

5.1 Cross-validation results

Table 4 reports the primary aggregate metrics across the three grouped, patient-level folds. Table 5 reports the corresponding per-fold values and confusion matrices, and Figure 4 visualizes

Table 4: Primary evaluation results, aggregated over three grouped cross-validation folds. Confidence intervals are 2000-resample bootstrap intervals over the three fold-level metric values; because the resample is drawn from only three numbers, these intervals mainly reflect the spread across folds rather than a high-resolution sampling distribution (see Section 7).

Metric	Mean	Std.	95% CI
Accuracy	88.66%	0.75%	[87.69%, 89.52%]
Balanced accuracy	88.26%	0.43%	[87.92%, 88.78%]
F1 (macro)	0.884	0.005	[0.877, 0.888]
AUC-ROC	0.970	0.003	[0.967, 0.973]
Cohen's κ	0.769	0.010	[0.755, 0.778]
MCC	0.777	0.011	[0.760, 0.786]

Table 5: Per-fold test results and confusion matrices (rows/columns ordered [non-tumor, tumor]).

Fold	n	Acc.	Bal. Acc.	F1	AUC	κ	MCC	Confusion matrix
1	65	87.69%	88.10%	0.877	0.967	0.755	0.761	$\begin{bmatrix} 28 & 2 \\ 6 & 29 \end{bmatrix}$
2	98	88.78%	88.78%	0.887	0.969	0.776	0.784	$\begin{bmatrix} 47 & 2 \\ 9 & 40 \end{bmatrix}$
3	105	89.52%	87.92%	0.888	0.973	0.778	0.786	$\begin{bmatrix} 60 & 2 \\ 9 & 34 \end{bmatrix}$

the spread of each metric across folds.

Performance is consistent across all three folds rather than driven by a single favorable split: the accuracy range is under 2 percentage points (87.69–89.52%) and the AUC-ROC range is under 0.007 (0.967–0.973). Table 6, computed directly from the three confusion matrices in Table 5, breaks this down into sensitivity, specificity, and predictive values. Sensitivity is lower and more variable across folds (79–83%) than specificity (93–97%), meaning the dominant error mode is missed tumor cases (false negatives) rather than false alarms. The model is comparatively conservative when uncertain, a pattern also visible in the confusion matrices (false positives are 2 in every fold, while false negatives range from 6 to 9).

5.2 Few-shot support size selection

Both $K = 2$ and $K = 8$ were trained and validated independently within each fold, and the better-validating configuration was carried forward to that fold's test evaluation. Table 7 shows the selected K per fold, recorded directly in the experiment log. Two of three folds preferred the smaller support size ($K = 2$); the model does not simply improve monotonically with more support examples, and the preferred K varies by fold. This is consistent with a genuinely small-data regime, where eight labeled query-adjacent examples do not reliably outperform two once the transferred CT prior is already carrying part of the classification signal.

5.3 Full-dataset inference

Beyond the three held-out fold evaluations, the saved pipeline also runs a single deployment-style inference pass with the best saved checkpoint over all 268 pathology samples. Table 8 reports this pass. We report it separately from Table 4 and do not treat it as an independent validation result, because the checkpoint used for this pass was trained on a subset of these same 268 samples (as part of one fold's training split); the 94.40% figure is therefore a deployment/consistency check on the saved pipeline, not a second held-out accuracy estimate.

No ROC or precision-recall curve was saved by the original pipeline for the cross-validation folds (only the scalar AUC-ROC in Table 4 was computed and stored). To give a visual sense of ranking behavior, Figure 5 shows an ROC and precision-recall curve that we computed ourselves, directly from the released full-dataset inference file, using the accompanying analysis

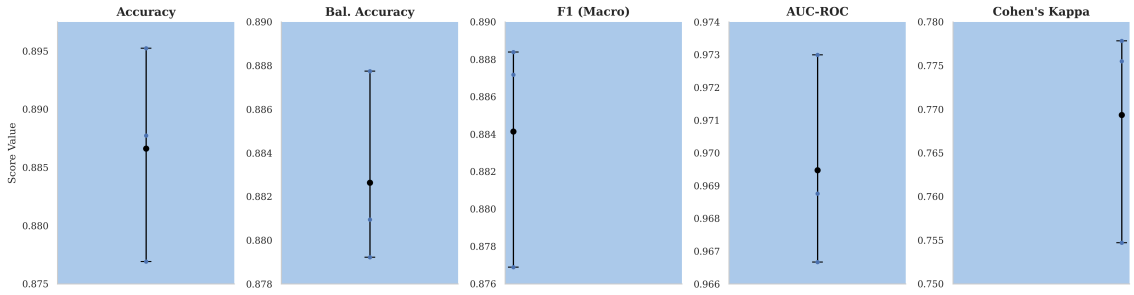


Figure 4: Range (min–max whiskers) and mean (marker) of each primary metric across the three grouped cross-validation folds. All metrics are tightly clustered, with AUC-ROC showing the smallest relative spread.

Table 6: Sensitivity, specificity, and predictive values, computed directly from the confusion matrices in Table 5 (positive class = tumor).

Fold	Sensitivity	Specificity	PPV	NPV
1	82.9%	93.3%	93.5%	82.4%
2	81.6%	95.9%	95.2%	83.9%
3	79.1%	96.8%	94.4%	87.0%
Mean	81.2%	95.3%	94.4%	84.4%

script.[3] Because this curve is computed on the same non-held-out inference pass as Table 8, its AUC (0.987) is optimistic relative to the validated cross-validation AUC of 0.970 in Table 4 and should be read only as an illustration of ranking shape, not as an additional validation result.

5.4 Calibration

Reliability diagrams were generated during Phase 2 training on each fold's held-out validation/test split. Figure 6(a) shows a representative diagram (fold checkpoint at epoch 20): the model's Expected Calibration Error (ECE) on this held-out slice is 0.290, and the reliability curve lies well below the diagonal at mid-to-high confidence: predictions with mean confidence near 0.85, for example, are correct only about half the time in this bin, well short of the 85% the confidence value would imply. This indicates the model is *overconfident* on genuinely unseen data, despite its strong ranking performance (AUC-ROC $\approx$ 0.97). None of the four training losses (classification, prototypical, adversarial, contrastive) directly penalizes miscalibration, so this is not unexpected, but it is an important caveat for any downstream use of the predicted probabilities as risk scores.

For contrast, Figure 6(b) shows the reliability diagram we computed on the full-dataset inference pass described in Table 8 (ECE = 0.025, using the released `tumor_probability` scores). This number looks much better, but for the same reason given above, it is computed on a pass where most samples were seen during training by the checkpoint being evaluated, so it reflects fit to already-seen data rather than generalization calibration. We include both panels specifically to make this distinction visible: **the held-out result (a) is the methodologically valid calibration estimate, and it indicates poor calibration.**

5.5 Training dynamics

Figure 7 shows training and validation accuracy over the 30 within-fold episodic training epochs (representative fold/K configuration). Training accuracy rises quickly and stabilizes near 0.95–

[3] `paper/scripts/make_result_figures.py` in the released repository.

Table 7: Selected few-shot support size per fold (from `experiment.log`).

Fold	Selected K	Validation accuracy at selection
1	2	0.972
2	8	0.944
3	2	0.778
Consensus (2 of 3 folds)		$K = 2$

Table 8: Full-dataset inference summary (not a held-out evaluation; see text).

Quantity	Value
Samples processed	268
Correct predictions	253 (94.40%)
Predicted tumor / non-tumor	124 / 144
Mean prediction confidence	0.969
Mean tumor probability (predicted-tumor subset)	0.976

1.0 within the first 5 epochs, while episodic validation accuracy is noisier and remains substantially lower, in roughly the 0.6–0.8 range, for the remainder of training. This gap is expected given the episodic setup: each validation point is computed on a small support/query episode rather than the full fold-level test set used for Table 5, so it is not directly comparable to the 87–90% fold-test accuracies reported above. It is a noisier, small-sample signal used only for checkpoint selection and early stopping during training.

5.6 Qualitative analysis of learned representations

Embedding structure. Figure 8 shows t-SNE projections of the 512-dimensional feature space at an early (epoch 4) and later (epoch 20) training checkpoint, colored by domain (CT vs. pathology) and by class. At both checkpoints, CT and pathology samples occupy visibly distinct regions of the projection – the two domains are not intermixed. Within the pathology region, however, samples separate into two class-consistent sub-clusters at both checkpoints, consistent with the strong classification performance reported above. We report this as a qualitative observation: the visualizations do not show the domains merging into a shared, intermixed space, which tempers any claim that adversarial alignment achieves full domain invariance at the representation level (Section 7). The classification signal in these plots appears to come primarily from class structure within the pathology domain rather than from a literal merging of the two modalities' embeddings.

Cross-dimensional attention. Figure 9 shows the attention weights assigned to the top-$k = 5$ retrieved CT context vectors (each one whole-volume descriptor from the 61-entry bank, not a sub-volume patch; Section 3) for a representative pathology query, at an early and a later training checkpoint. In both cases the weights are not perfectly uniform (they range approximately 0.17–0.23 around the uniform value of 0.20), indicating the model applies some differential weighting to the retrieved CT volumes rather than averaging all k identically. We did not compute a quantitative selectivity statistic (e.g., attention entropy) for this run and did not observe a dramatic sharpening between checkpoints in the saved images, so we report this as a mild, qualitative signal that the retrieval-and-attention path is active, rather than as a quantified trend.

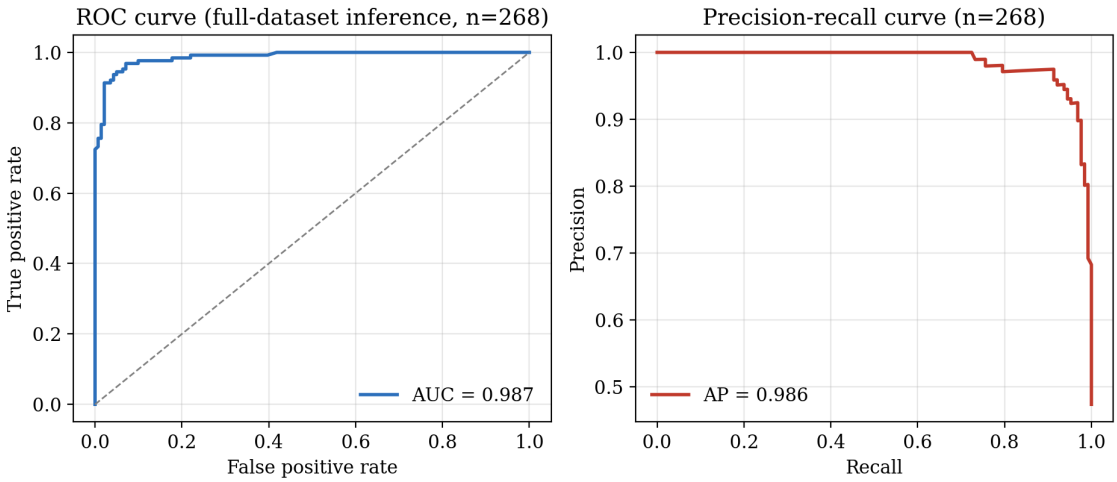


Figure 5: ROC and precision-recall curves computed from the full-dataset inference output (268 samples; not a held-out split, see text). AUC and average precision (AP) here are not directly comparable to the cross-validation AUC-ROC in Table 4.

Prototype evolution. Figure 10 tracks the L2 distance each class prototype moves between consecutive checkpoints over the course of training. Both class prototypes drift substantially over the first ∼20 checkpoints (from an initial value near 10–14 down to a stable band), after which drift settles into a lower, roughly stationary oscillation for the remainder of training. Class 0 and Class 1 prototypes show closely similar drift magnitude throughout, with no indication of prototype collapse (drift never approaches zero for either class).

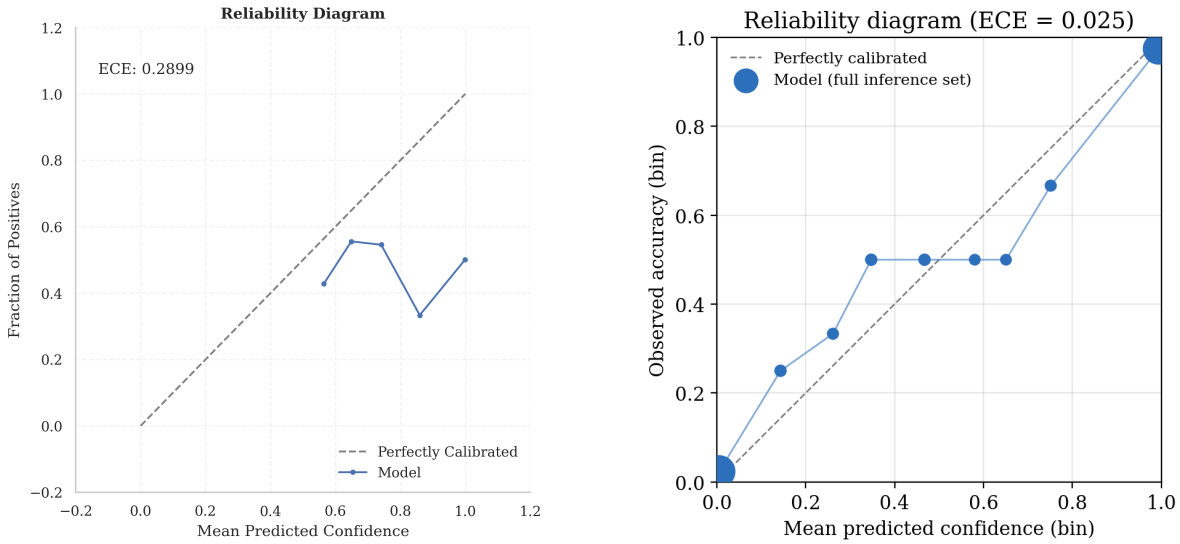
6 Discussion

6.1 Does CT-derived structure transfer to pathology classification?

The central question of this paper was whether self-supervised structure learned from 3D CT volumes can support few-shot classification of 2D pathology images. The results in Section 5.1 are consistent with a positive answer at the level of a single, carefully validated domain pair: the framework reaches 0.970 mean AUC-ROC and 88.7% mean accuracy under $K \in \{2, 8\}$-shot supervision, with narrow spread across three grouped, patient-level folds. Because the source and target domains share no paired samples, no common patients, and no common anatomical site, any signal the model exploits has to be a generic structural regularity rather than a shortcut specific to this dataset pairing (e.g., scanner artifacts correlated with labels). We treat this as evidence of feasibility, not as evidence that the specific gain is attributable to any one architectural component. That attribution requires the controlled ablations we did not run (Section 7).

6.2 A conservative decision boundary

The confusion matrices in Table 5 show a consistent asymmetry: false positives are rare (2 in every fold) while false negatives are more common (6–9 per fold), giving a mean sensitivity of 81.2% against a mean specificity of 95.3% (Table 6). In a screening context this pattern is a genuine concern, since missed tumor cases are typically more costly than false alarms. It suggests that if this framework were used as anything beyond a research prototype, it would need either a lowered decision threshold (trading some specificity for sensitivity) or a human-in-the-loop review step for negative predictions with borderline confidence.



(a) Held-out fold validation/test split

(b) Full-dataset inference (not held out)

Figure 6: Reliability diagrams under two evaluation conditions. On a genuinely held-out split (a), the model is markedly overconfident (ECE = 0.290). On the non-held-out full-dataset inference pass (b), apparent calibration is much better (ECE = 0.025), which we attribute to train/inference overlap rather than genuine reliability; see Section 5.4.

6.3 Few-shot support size is not monotonic

Two of three folds selected the smaller support size ($K = 2$) over $K = 8$ (Table 7). If the transferred CT context were doing little work, we would expect performance to scale more predictably with additional labeled support, since the pathology-only signal would then dominate. The fact that $K = 2$ is competitive with, and in two folds preferred over, $K = 8$ is consistent with (though does not by itself prove) the retrieved CT context substituting for some of the information that additional labeled examples would otherwise provide.

6.4 Strong discrimination, poor calibration

The most important nuance in our results is the gap between discrimination and calibration. AUC-ROC of 0.97 indicates the model ranks tumor and non-tumor samples very effectively across thresholds. But the held-out reliability diagram (Figure 6a) shows an Expected Calibration Error of 0.29, with predicted confidence substantially exceeding observed accuracy in the mid-to-high confidence range. This is not a contradiction: a model can separate two classes well while still reporting probabilities that do not reflect true likelihoods, because ranking quality (AUC) and calibration are different properties, and nothing in our training objective (classification, prototypical, adversarial, or contrastive loss) directly penalizes miscalibration. This finding matters for any reading of this framework as clinically relevant: the class predictions and their relative ranking appear trustworthy, but the raw probability values should not be interpreted as calibrated risk estimates without post-hoc correction (e.g., temperature scaling or Platt scaling on a held-out calibration set), which we did not implement in this study.

6.5 Partial, not complete, representation-level alignment

The t-SNE visualizations in Figure 8 show pathology samples separating cleanly by class, which is consistent with the classification performance, but they also show CT and pathology samples remaining in visually distinct regions of the projection at both the early and later checkpoints examined. This means we cannot claim that adversarial alignment collapses the two domains

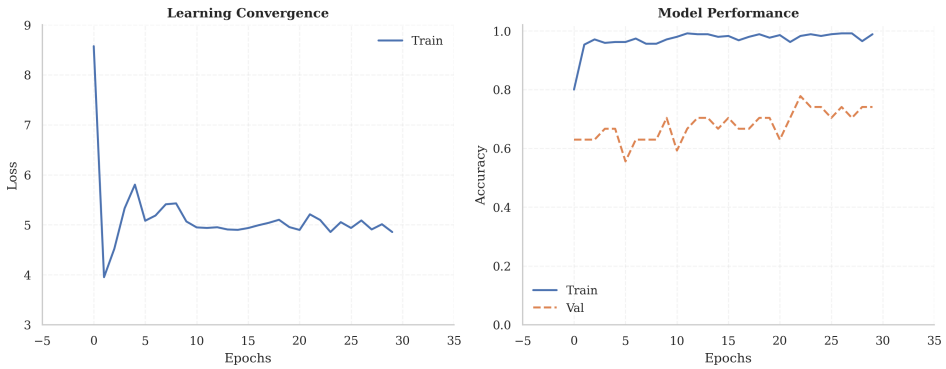
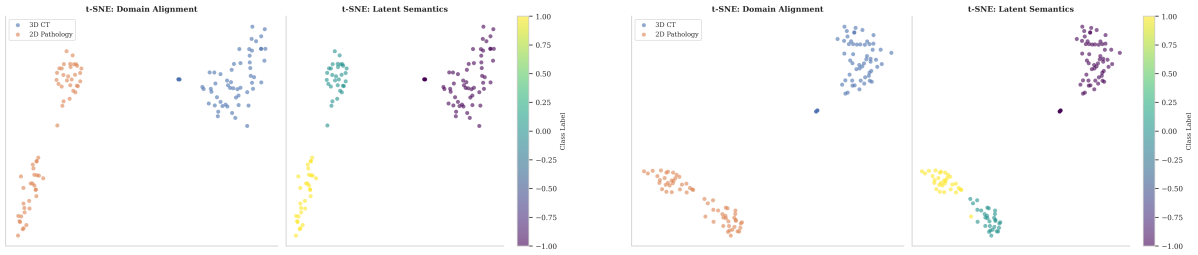


Figure 7: Training loss and episodic train/validation accuracy over 30 epochs of Phase 2 training (representative run). Validation accuracy is computed per-episode and is noisier and lower than the fold-level test accuracy in Table 5.



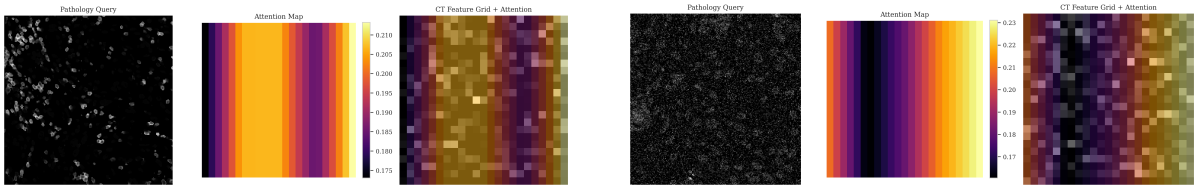
(a) Epoch 4 **(b)** Epoch 20

Figure 8: t-SNE projections of the shared feature space, colored by domain (left half of each panel) and by class label (right half). CT and pathology samples remain visually separated by domain at both checkpoints; pathology samples separate into two class-consistent groups at both checkpoints.

into a shared, indistinguishable representation. Two explanations are consistent with this observation and with the classification performance still being strong. First, the retrieval-and-attention mechanism does not require CT and pathology features to occupy the same region of representation space: it only requires that cosine similarity between a pathology query and the 61-entry CT bank surface whole-volume descriptors that are useful once passed through the learned cross-attention and gating, which is a weaker requirement than full domain invariance. Second, a 2-dimensional t-SNE projection can fail to reveal alignment that exists in the full 512-dimensional space; t-SNE preserves local neighborhood structure but not global geometry, so this observation is suggestive rather than conclusive. Either way, we do not repeat the stronger claim from the underlying thesis that the two domains become intermixed, since the visualizations we inspected do not support it.

6.6 Failure pattern

Consistent with Section 5.1, the dominant error mode across all three folds is false negatives. This suggests the most difficult cases are pathology samples with a weak or ambiguous tumor signal, plausibly ones near the 5% tumor-mask-ratio labeling threshold used to construct the weak labels, where the ground truth itself is least certain. This is a property of the task's label construction as much as of the model, and it is a natural place for the weak-label auditing proposed in Section 8.



(a) Early checkpoint **(b)** Later checkpoint

Figure 9: Attention weights over the top-$k = 5$ retrieved CT volume-level descriptors for a representative pathology query. Weights deviate mildly from uniform ($\pm 15\%$ of the uniform value $1/k = 0.20$) at both checkpoints shown.

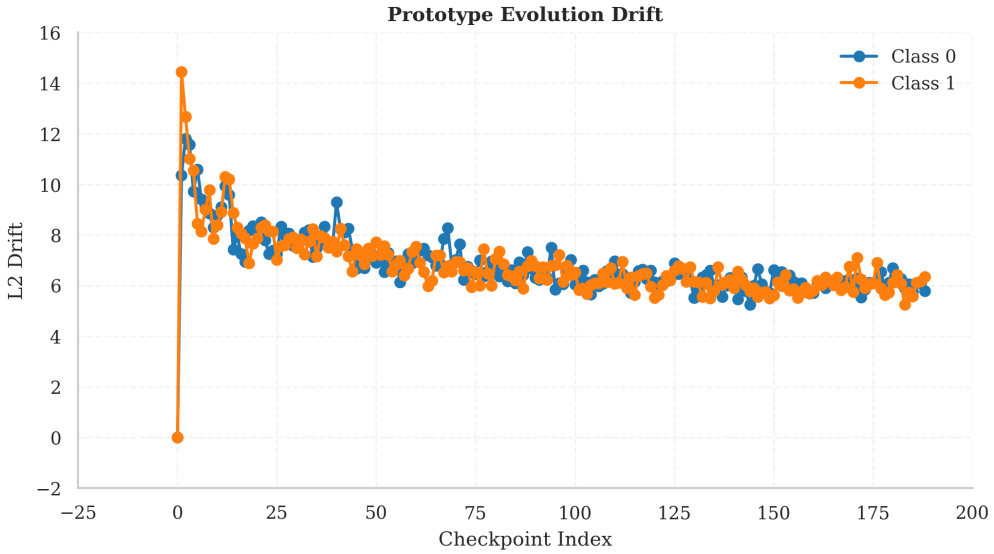


Figure 10: Per-checkpoint L2 drift of the two class prototypes over training. Both prototypes stabilize into a lower, roughly stationary drift band after an initial adjustment period.

6.7 Comparison with prior work

We do not attempt a numerical comparison with prior few-shot medical imaging or domain adaptation results, because no directly comparable published benchmark exists for 3D-CT-to-2D-pathology transfer on this dataset pair: differences in task definition, label source, and evaluation protocol would make any such comparison more misleading than informative (Section 2). The comparison we can make is methodological. The design evaluated here uses grouped, patient-level splits, reports multiple reliability-oriented metrics rather than accuracy alone, and explicitly separates a held-out cross-validation estimate from a non-held-out deployment-style inference pass; these practices are not universal in the few-shot medical imaging literature we reviewed.

7 Limitations

We list concrete limitations of the present study, distinguishing what was implemented and evaluated from what was not, so that the contribution above is not read more broadly than the evidence supports.

No component ablation was run. The released codebase includes classes for removing or replacing individual components (the attention bridge, adversarial alignment, and prototypical

branch) and for a classifier-only baseline, but these variants were not trained or evaluated as part of the reported run: only the full model was trained and tested on the grouped folds. We therefore cannot attribute the reported performance to any specific component, and we do not report an ablation table. This is the most important open question for follow-up work (Section 8). It also means we cannot currently rule out that the reported gains come from the pathology encoder and episodic training alone, with the CT-transfer machinery contributing little; a pathology-only baseline and per-component ablations (no CDAM, no adversarial alignment, no InfoNCE, retrieval from a shuffled or random CT bank, $k = 0$) are needed before that attribution can be made with confidence.

The contrastive alignment loss has a circularity risk. As detailed in Section 3, $\mathcal{L}_{\text{ctr}}$ pairs each pathology query's projection with the projection of $\mathbf{r}$, the retrieval-weighted CT summary computed *from that same query*, rather than with an independently sourced positive example. This is a structural consequence of there being no ground-truth CT-pathology correspondence to train against, but it means the loss may reinforce self-consistency with the retrieval mechanism's own similarity ranking rather than learning an externally meaningful alignment. We are not aware of a way to fully resolve this without either a differently-designed contrastive objective or an external validation signal for retrieval quality, and we did not implement either in this study.

The gate floor and usage penalty force transfer usage rather than testing for it. The transfer gate is bounded away from zero (floor 0.15) and the usage-penalty loss (Eq. (11)) explicitly discourages the fused representation from staying close to the untransferred pathology features. Both choices were made to keep the transfer pathway numerically active during training, but they also mean the model cannot freely learn to ignore unhelpful CT context if that were the better strategy for a given fold or sample. The reported performance therefore reflects a model that is structurally biased toward using CT context, not one that discovered independently that doing so helps; only an ablation with the floor and penalty removed could distinguish forced use from beneficial use.

No controlled statistical significance testing was run. Permutation testing and standardized effect-size comparisons (e.g., against a trained baseline) were not executed in this study. The confidence intervals we report (Table 4) are bootstrap intervals computed by resampling, with replacement, from only the three fold-level metric values. With $n = 3$, the resample can only draw from three numbers, so the resulting interval mainly redescribes the spread across folds rather than providing the higher-resolution uncertainty estimate that bootstrapping over per-sample predictions would give. We report it because it is what the released pipeline computes, but it should not be read as a tight, high-power confidence interval, and it is not a substitute for a significance test against an alternative model.

No systematic robustness testing was run. We did not evaluate the model's sensitivity to injected input noise or to missing/dropped pathology channels in this study. Given the model's channel-attention mechanism, such a test would be informative future work.

Calibration is weak on held-out data. As discussed in Section 5.4 and Section 6, the model is measurably overconfident on genuinely held-out validation/test data (ECE ≈ 0.29), even though it discriminates well (AUC-ROC ≈ 0.97). We did not apply any post-hoc calibration correction. Any use of the predicted probabilities as risk scores, rather than as inputs to thresholded classification, should account for this.

Representation-level domain alignment is partial, and only assessed qualitatively. The t-SNE visualizations available to us (Figure 8) show CT and pathology features remaining visually separated by domain; we do not have evidence that adversarial alignment achieves full domain invariance at the representation level, only that classification performance is strong despite this partial alignment (Section 6). We also did not compute any quantitative alignment measure (for example, maximum mean discrepancy between the two feature distributions, or the accuracy of a linear probe trained to classify domain from the frozen fused features) that would corroborate or contradict the t-SNE impression with a number rather than a picture. t-SNE preserves local neighborhood structure but not global geometry, so a numerical check would be a meaningfully stronger form of evidence than what we report here.

Few-shot support size selection used only two candidate values. Within each fold, $K = 2$ and $K = 8$ were each trained to convergence and the better-validating configuration (chosen on that fold's own validation split, never its test split) was carried forward to test evaluation (Table 7); this is standard validation-based model selection and does not leak test-set information. It does not, however, characterize how performance varies over a denser range of K (e.g., 1, 3, 5, 10) or over the retrieval hyperparameters k and τ, and we did not run a nested cross-validation or repeated-sweep procedure to quantify selection variance across a larger candidate set.

Single domain pair, single seed. All experiments use one source domain (lung CT), one target domain (HNSCC pathology), and one random seed (42). We do not have a multi-seed variance estimate for the reported metrics – the reported standard deviations and confidence intervals characterize variability across the three grouped folds under this one seed, not run-to-run stochasticity from re-initialization or re-shuffling. Hyperparameters (Table 3) were set through implementation-driven tuning and validation monitoring rather than a systematic search, so we cannot rule out that a different configuration would perform differently.

Weak, threshold-derived labels. Target-domain labels are derived from a 5% tumor-mask-ratio threshold rather than from a dedicated expert classification campaign. Samples near this threshold carry inherent label ambiguity, which likely contributes to the false-negative-dominated error pattern discussed in Section 6.

Small target dataset. The target dataset contains 268 samples from 42 patients. This is an appropriate scale for a few-shot evaluation but is small relative to what would be needed to support a general clinical generalization claim.

Full-dataset inference is not an independent validation. The 94.40% accuracy reported in Table 8 comes from running a single saved checkpoint over all 268 samples, most of which were seen during that checkpoint's own fold training. It is a pipeline consistency check, not a second held-out performance estimate, and should not be averaged or compared directly with the cross-validation numbers in Table 4.

Untested generalization boundaries. We have not tested the framework under a different CT acquisition protocol, a different pathology staining protocol, a different institution, or a different cancer type. Performance under such shifts is unknown.

8 Future Work

Ablations and baselines. The most direct next step is a controlled ablation study, run on the same grouped folds used here: removing or replacing CDAM, the adversarial discriminator,

and the prototypical branch one at a time, and comparing against a pathology-only baseline trained without any CT transfer. We did not run these controlled comparisons in the present study (Section 7) and treat them as the immediate priority for follow-up work, since they would let us attribute the observed performance to specific components rather than to the framework as a whole.

Calibration correction. Given the held-out overconfidence documented in Section 5.4, a natural next step is to apply and evaluate post-hoc calibration methods (e.g., temperature scaling or Platt scaling fit on a held-out calibration split) and to measure whether they close the gap between predicted confidence and observed accuracy without degrading discrimination.

Additional domain pairs. The framework should be tested on other radiology-to-pathology combinations, for example brain MRI to neuropathology, breast MRI to histopathology, or abdominal CT to gastrointestinal pathology, to establish whether the transfer benefit generalizes beyond the lung-CT-to-HNSCC pair studied here, or whether it depends on incidental similarities between this particular source and target.

Stronger and externally-validated labels. The target task uses weak labels derived from tumor-mask ratio. Replacing or auditing these with expert-assigned classification labels, and validating on an independent HNSCC cohort from a different institution, scanner, or staining protocol, would test whether the reported metrics hold outside the single dataset used here.

Additional source modalities. Extending the CT feature bank to incorporate other volumetric or cross-sectional modalities (e.g., MRI, PET) as additional or alternative sources could test whether the retrieval-and-fusion mechanism generalizes across source modalities, not only across the specific CT acquisition protocol used in LungCT-Diagnosis.

Stronger pretrained backbones. The CT encoder here is trained from scratch with 10 epochs of in-house contrastive self-supervision on 61 volumes, and the pathology encoder is likewise trained from scratch rather than initialized from a larger pretrained model. Substituting either encoder for a larger, externally pretrained CT or pathology foundation model, if a suitable and appropriately licensed one is identified, would test whether the transfer benefit reported here is limited by the weak CT prior, the weak pathology prior, or the transfer mechanism itself. We leave specific candidates unnamed here rather than cite them without having independently verified them.

Quantitative alignment diagnostics. Beyond the qualitative t-SNE inspection in Section 5.6, future evaluations should report a quantitative domain-alignment measure, such as maximum mean discrepancy between CT and pathology feature distributions or the accuracy of a linear probe trained to predict domain from frozen fused features, and retrieval-quality diagnostics such as the similarity-score distribution of retrieved versus random CT entries and qualitative inspection of which CT volumes are retrieved for tumor versus non-tumor pathology queries.

Deployment-oriented engineering. The current pipeline requires 3D CT encoding and feature-bank construction, which adds computational overhead relative to a pathology-only classifier. Precomputing and shipping a fixed CT feature bank, and optimizing the pathology-side inference path independently of the (frozen) CT encoder, would clarify the actual deployment cost of the transfer mechanism relative to its benefit.

Multi-modal and language-integrated extensions. Combining the visual transfer mechanism studied here with clinical text (e.g., radiology or pathology reports) or with large language models is a natural extension, potentially enabling diagnostic support systems that reason jointly over volumetric imaging, planar pathology, and narrative clinical information. We flag this as a direction rather than a claim, since it was not implemented or tested in this study.

9 Conclusion

We asked whether self-supervised structure learned from abundant, unlabeled 3D CT volumes can be transferred to improve few-shot classification of scarce, weakly-labeled 2D pathology images, given that the two domains differ in dimensionality, modality, patient population, and anatomical site. We built a framework that pretrains a volumetric encoder with contrastive self-supervision, retrieves relevant CT context through top-k similarity search, fuses that context into pathology features through a cross-dimensional attention bridge with gated residual connections, aligns the two feature spaces adversarially, and classifies pathology samples episodically through a prototypical network. Evaluated with grouped, patient-level three-fold cross-validation on 268 HNSCC pathology samples under $K \in \{2, 8\}$-shot supervision, the framework reached a mean accuracy of 88.7% and a mean AUC-ROC of 0.970, with consistent metrics across all three folds and a full-dataset inference accuracy of 94.4% (on a non-held-out pass). The picture is not uniformly positive: held-out calibration is poor despite strong discrimination, and embedding visualizations show only partial representation-level alignment between the two domains. We report both the positive and the negative findings, because an honest account of what did and did not work is more useful to future work on this problem than a selectively positive one.

The result is a working demonstration, on one public radiology-pathology pair, that cross-dimensional transfer is technically feasible and can support a data-efficient pathology classifier without a large target-domain annotation campaign. It is not evidence that this specific source-target pairing, weak-label formulation, or architecture generalizes beyond the setting tested here – that requires the controlled ablations, additional domain pairs, and external validation outlined in Section 8. We release the full training and evaluation pipeline, hyperparameter configuration, and generated figures to support that follow-up work.

Data and Code Availability

Both datasets are publicly available through The Cancer Imaging Archive (TCIA) under standard TCIA data-use terms: LungCT-Diagnosis[4] and HNSCC-mIF-mIHC-Comparison.[5] Both collections are de-identified in accordance with HIPAA Safe Harbor requirements; no additional ethical approval was required for this secondary analysis of de-identified public data. The training and evaluation code, hyperparameter configuration, grouped cross-validation splits (patient IDs per fold), and saved result artifacts (per-fold metrics, predictions, and figures) are available at `https://github.com/zulqarnainalipk`. Trained model checkpoints are not included in that repository, to keep it lightweight; they can be regenerated by re-running the released code with the provided splits and seed, or obtained from the author on request.

[4] `https://www.cancerimagingarchive.net/collection/lungct-diagnosis/`, DOI: `10.7937/K9/TCIA.2015.A6V7JIWX`.

[5] `https://www.cancerimagingarchive.net/collection/hnscc-mif-mihc-comparison/`, DOI: `10.7937/TCIA.2020.T90F-WB82`.

References

[1] Eva Pachetti and Sara Colantonio. A systematic review of few-shot learning in medical imaging. *arXiv preprint arXiv:2309.11433*, 2023.

[2] Jayanta Nayem, Sk Shalauddin Hasan, Nasmin Amina, Bishmoy Das, Md. Solaiman Ali, Munshi Mahbubur Ahsan, and Shivsubramani Raman. Few shot learning for medical imaging: A comparative analysis of methodologies and formal mathematical framework. *arXiv preprint arXiv:2305.04401*, 2023.

[3] Hao Guan and Mingxia Liu. Domain adaptation for medical image analysis: A survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2022.

[4] Yuemeng Li and Yong Fan. Medical image segmentation with domain adaptation: A survey. *arXiv preprint arXiv:2311.01702*, 2023.

[5] Lin Zhao, Xiao Chen, Eric Z. Chen, Yikang Liu, Terrence Chen, and Shanhui Sun. Retrieval-augmented few-shot medical image segmentation with foundation models. *arXiv preprint arXiv:2408.08813*, 2024.

[6] Kai He, Chen Gan, Zhuoyuan Li, Islem Rekik, Zihao Yin, Wen Ji, Yang Gao, Qian Wang, Jun Zhang, and Dinggang Shen. Transformers in medical image analysis: A review. *arXiv preprint arXiv:2202.12165*, 2022.

[7] Reza Azad, Amirhossein Kazerouni, Moein Heidari, Ehsan Khodapanah Aghdam, Amirali Molaei, Yiwei Jia, Abin Jose, Rijo Roy, and Dorit Merhof. Advances in medical image analysis with vision transformers: A comprehensive review. *arXiv preprint arXiv:2301.03505*, 2023.

[8] Wei-Chien Wang, Euijoon Ahn, Dagan Feng, and Jinman Kim. A review of predictive and contrastive self-supervised learning for medical images. *arXiv preprint arXiv:2302.05043*, 2023.

[9] Magdalini Paschali, Zhihong Chen, Louis Blankemeier, Maya Varma, Alaa Youssef, Christian Bluethgen, Curtis Langlotz, Sergios Gatidis, and Akshay Chaudhari. Foundation models in radiology: What, how, when, why and why not. *arXiv preprint arXiv:2411.18730*, 2024.

[10] Zifeng Wang, Zhenbang Wu, Dinesh Agarwal, and Jimeng Sun. MedCLIP: Contrastive learning from unpaired medical images and text. *arXiv preprint arXiv:2210.10163*, 2022.

[11] Qingqiu Han, Jing Liu, Zichen Qin, and Zhiming Zheng. Integrating MedCLIP and cross-modal fusion for automatic radiology report generation. *arXiv preprint arXiv:2412.07141*, 2024.

[12] Yitian Tao, Liyan Ma, Jing Yu, and Huaqi Zhang. Memory-based cross-modal semantic alignment network for radiology report generation. *arXiv preprint arXiv:2404.00588*, 2024.

[13] Jun Wang, Lixing Zhu, Xiaohao Yu, Abhir Bhalerao, and Yulan He. Improving medical visual representation learning with pathological-level cross-modal alignment and correlation exploration. *arXiv preprint arXiv:2506.10573*, 2025.

[14] Juan Manuel Zambrano Chaves, Shih-Cheng Huang, Yanbo Xu, Hanwen Xu, Naoto Usuyama, Sheng Zhang, Fei Wang, Yujia Xie, Mahmoud Khademi, Ziyi Yang, Hany Awadalla, Julia Gong, Houdong Hu, Jianwei Yang, Chunyuan Li, Jianfeng Gao, Yu Gu,

Cliff Wong, Mu Wei, Tristan Naumann, Muhao Chen, Matthew P. Lungren, Akshay Chaudhari, Serena Yeung-Levy, Curtis P. Langlotz, Sheng Wang, and Hoifung Poon. Towards a clinically accessible radiology foundation model: Open-access and lightweight, with automated evaluation. *arXiv preprint arXiv:2403.08002*, 2024.

[15] Xiaoman Liu, Guang Yang, Yuan Luo, Jiancheng Mao, Xiao Zhang, Mingyuan Gao, Shu Zhang, Jun Shen, and Guang Wang. Expert-level vision-language foundation model for real-world radiology and comprehensive evaluation. *arXiv preprint arXiv:2409.16183*, 2024.

[16] Yuan Tian, Zhiliang Xu, Yang Ma, Wanli Ding, Ruixuan Wang, Zhifan Gao, Guang Cheng, Liang He, and Xiaoying Zhao. Survey on deep learning in multimodal medical imaging for cancer detection. *arXiv preprint arXiv:2312.01573*, 2023.

[17] Callum Jennings, Alexander Broad, Lucy Godson, Erika Clarke, David Westhead, and Darren Treanor. Machine learning-based multimodal prognostic models integrating pathology images and high-throughput omic data for overall survival prediction in cancer: A systematic review. *arXiv preprint arXiv:2507.16876*, 2025.

[18] Dingyi Li, Guangyao Wan, Xu Wu, Xiaoming Wu, Xi Chen, Yun He, Christine G. Lian, Peter K. Sorger, Yevgeniy R. Semenov, and Chen Zhao. Multi-modal foundation models for computational pathology: A survey. *arXiv preprint arXiv:2503.09091*, 2025.

[19] Maximilian Nielsen, Laura Wenderoth, Thilo Sentker, and Rene Werner. Self-supervision for medical image classification: State-of-the-art performance with about 100 labeled training samples per class. *arXiv preprint arXiv:2304.05163*, 2023.

[20] Tim Lenz, Omar S. M. El Nahhas, Marta Ligero, and Jakob Nikolas Kather. Reducing self-supervised learning complexity improves weakly-supervised classification performance in computational pathology. *arXiv preprint arXiv:2403.04558*, 2024.

[21] Peter Neidlinger, Omar S. M. El Nahhas, Hannah Sophie Muti, Tim Lenz, Michael Hoffmeister, Hermann Brenner, Marko van Treeck, Rupert Langer, Bastian Dislich, Hans Michael Behrens, Christoph Röcken, Sebastian Foersch, Daniel Truhn, Antonio Marra, Oliveira L. Saldanha, and Jakob Nikolas Kather. Benchmarking foundation models as feature extractors for weakly-supervised computational pathology. *arXiv preprint arXiv:2408.15823*, 2024.

[22] Dingyi Li, Guangyao Wan, Xu Wu, Xiaoming Wu, Aashish Jain Nirmal, Christine G. Lian, Peter K. Sorger, Yevgeniy R. Semenov, and Chen Zhao. A survey on computational pathology foundation models: Datasets, adaptation strategies, and evaluation tasks. *arXiv preprint arXiv:2501.15724*, 2025.

[23] Omar S. M. El Nahhas, Georg Wölflein, Marta Ligero, Tim Lenz, Marko van Treeck, Firas Khader, Daniel Truhn, and Jakob Nikolas Kather. Joint multi-task learning improves weakly-supervised biomarker prediction in computational pathology. *arXiv preprint arXiv:2403.03891*, 2024.

[24] Abhinav Sharma, Bojing Liu, and Mattias Rantalainen. WEEP: A method for spatial interpretation of weakly supervised CNN models in computational pathology. *arXiv preprint arXiv:2403.15238*, 2024.

[25] Jun Ma, Yuting He, Feifei Li, Lin Han, Chenyu You, and Bo Wang. Segment anything in medical images. *arXiv preprint arXiv:2304.12306*, 2023.

[26] Saikat Roy, Tassilo Wald, Gregor Koehler, Maximilian R. Rokuss, Nico Disch, Julius Holzschuh, David Zimmerer, and Klaus H. Maier-Hein. SAM.MD: Zero-shot medical image segmentation capabilities of the segment anything model. *arXiv preprint arXiv:2304.05396*, 2023.

[27] Peilun Shi, Jianing Qiu, Sai Mu Dalike Abaxi, Hao Wei, Frank Po-Wen Lo, and Wu Yuan. Generalist vision foundation models for medical imaging: A case study of segment anything model on zero-shot medical segmentation. *arXiv preprint arXiv:2304.12637*, 2023.

[28] Junde Wu, Wei Ji, Yuanpei Liu, Huazhu Fu, Min Xu, Yueming Xu, and Yingying Jin. Medical SAM adapter: Adapting segment anything model for medical image segmentation. *arXiv preprint arXiv:2304.12620*, 2023.

[29] Xin Wei, Jiaxin Cao, Yizhe Jin, Mengyao Lu, Guohua Wang, and Shuo Zhang. I-MedSAM: Implicit medical image segmentation with segment anything. *arXiv preprint arXiv:2311.17081*, 2023.

[30] Ali Hatamizadeh, Vishwesh Nath, Yucheng Tang, Dong Yang, Holger R. Roth, and Daguang Xu. Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images. *arXiv preprint arXiv:2201.01266*, 2022.

[31] Fabian Isensee, Tassilo Wald, Constantin Ulrich, Michael Baumgartner, Saikat Roy, Klaus Maier-Hein, and Paul F. Jaeger. nnU-Net revisited: A call for rigorous validation in 3D medical image segmentation. *arXiv preprint arXiv:2404.09556*, 2024.

[32] Nikhil Dhariwal and Aditya Giridharan. Towards precision oncology: Predicting mortality and relapse-free survival in head and neck cancer using clinical data. *arXiv preprint arXiv:2502.11200*, 2025.